

DEF

Danakil Depression, Ethiopia 123
 darmstadtium 67
 Dead Sea 131
 deuterium 10
 diamond 11, 104
 Döbereiner, Johann 12
 dubnium 65
 dysprosium 77
 earth, classical element 4
 “earths” 31, 32, 70
 Earth’s atmosphere 5, 112, 127
 Einstein, Albert 89
 einsteinium 89
 electron shells 9
 electrons 5, 8, 9
 erbium 70, 79
 europium 71, 74–75
 ferberite 58
 fermium 90
 fire 4, 112, 122
 fission 11, 66
 flames 22, 33
 Flerov, Georgy 108
 flerovium 108
 fluorine 132, 134
 fluorite 134
 francium 28–29, 137

GHI

Gadolin, Johann 75
 gadolinite 70, 75, 77
 gadolinium 75
 galena 53
 galinstan 98
 gallium 98
 gamma rays 11
 garnierite 50
 gases 6, 112, 138–145
 see also non-metals
 germanium 103, 108

Ghiorso, Albert 90
 glass 100–101
 gold 5, 6, 40, 61, 62–63
 graphite 11, 103
 greenockite 57
 groups (periodic table) 15
 gypsum 35
 hafnium 57
 halite 132, 135
 halogen group 130–137
 hard metals 40
 hassium 66
 helium 18, 88, 140, 141, 142
 holmium 77
 human body 5, 102, 112,
 132
 hydrochloric acid 133
 hydrogen 8, 10, 16–19, 124,
 141
 and calcium 35
 and halogens 132
 and methane 107
 bomb (H-bomb) 89, 90
 indium 95, 99
 inert gases 141
 insulators 95, 103
 iodine 131, 132–133, 136
 iridium 59
 iron 41, 46
 and corrosion 123
 in alloys 34, 44
 in compounds 48–49
 in magnets 50, 71, 74
 isotopes 10, 11, 87

JKL

jewellery 40, 54, 56, 61
 Joint Institute for Nuclear
 Research (JINR), Russia 65,
 119, 129
 Karymsky volcano, Russia 113

kernite 96
 kohl 118
 krypton 144
 Lake Baikal, Russia 106–107
 Lake Natron, Tanzania 26–27
 lanthanides 14, 68–79
 lanthanum 68–69, 72
 Lavoisier, Antoine 12
 Lawrence, Ernest 91
 Lawrence Berkeley National
 Laboratory, USA 83
 lawrencium 91
 lead 53, 67, 108, 109, 119
 lepidolite 24
 leucite 28
 light-emitting diodes (LEDs) 71
 liquids 6
 lithium 8, 12, 22, 24
 Livermore, USA 129
 livermorium 122, 129
 living organisms 102, 121
 see also human body
 lutetium 79

M

magnesite 32
 magnesium 32, 33, 34
 magnetic elements
 lanthanides 71, 74, 76
 transition metals 47, 50, 53
 malachite 7
 manganese 45
 matter *see* gases; liquids; solids
 Meitner, Lise 66
 meitnerium 66
 melting points 32, 94, 98, 119
 of transition metals 40,
 58, 60
 Mendeleev, Dmitri 13, 90
 mendelevium 90
 mercury 6, 12, 64, 76